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THREE-AXIS TUNED MASS DAMPER FOR TURBO MOLECULAR PUMP

Abstract

A mass damper is connected to a vacuum pump to reduce vibrations during operation. The mass damper includes a conic portion defined by a converging end and a diverging end and a damping mass connected to the converging end of the conic portion. The vacuum pump has a pump body, a vacuum inlet in fluid communication with an internal volume of a vacuum chamber, a vacuum outlet, and a rotor assembly disposed within the pump body. The rotor assembly is configured to rotate about a rotary axis to pump gas from the internal volume of the vacuum chamber out of the vacuum outlet. The conic portion may have a cone axis that is coaxial with the rotary axis of the vacuum pump.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims priority to the provisional patent application filed Feb. 28, 2024, and assigned U.S. App. No. 63/558,662, the disclosure of which is hereby incorporated herein by reference.

FIELD OF THE DISCLOSURE

[0002] This disclosure relates to damping systems for vacuum pumps used in semiconductor manufacturing.

BACKGROUND OF THE DISCLOSURE

[0003] Evolution of the semiconductor manufacturing industry is placing greater demands on yield management and, in particular, on metrology and inspection systems. Critical dimensions continue to shrink, yet the industry needs to decrease time for achieving high-yield, high-value production. Minimizing the total time from detecting a yield problem to fixing it determines the return-on-investment for a semiconductor manufacturer.

[0004] Fabricating semiconductor devices, such as logic and memory devices, typically includes processing a workpiece, such as a semiconductor wafer, using a large number of fabrication processes to form various features and multiple levels of the semiconductor devices. For example, lithography is a semiconductor fabrication process that involves transferring a pattern from a reticle to a photoresist arranged on a semiconductor wafer. Additional examples of semiconductor fabrication processes include, but are not limited to, chemical-mechanical polishing (CMP), etch, deposition, and ion implantation. An arrangement of multiple semiconductor devices fabricated on a single semiconductor wafer may be separated into individual semiconductor devices.

[0005] Inspection processes are used at various steps during semiconductor manufacturing to detect defects on wafers to promote higher yield in the manufacturing process and, thus, higher profits. Inspection has always been an important part of fabricating semiconductor devices such as integrated circuits (ICs). However, as the dimensions of semiconductor devices decrease, inspection becomes even more important to the successful manufacture of acceptable semiconductor devices because smaller defects can cause the devices to fail. For instance, as the dimensions of semiconductor devices decrease, detection of defects of decreasing size has become necessary because even relatively small defects may cause unwanted aberrations in the semiconductor devices.

[0006] As design rules shrink, however, semiconductor manufacturing processes may be operating closer to the limitation on the performance capability of the processes. In addition, smaller defects can have an impact on the electrical parameters of the device as the design rules shrink, which drives more sensitive inspections. As design rules shrink, the population of potentially yield-relevant defects detected by inspection grows dramatically, and the population of nuisance defects detected by inspection also increases dramatically. Therefore, more defects may be detected on the wafers, and correcting the processes to eliminate all of the defects may be difficult and expensive. Determining which of the defects actually have an effect on the electrical parameters of the devices and the yield may allow process control methods to be focused on those defects while largely ignoring others. Furthermore, at smaller design rules, process-induced failures, in some cases, tend to be systematic. That is, process-induced failures tend to fail at predetermined design patterns often repeated many times within the design. Elimination of spatially-systematic, electrically-relevant defects can have an impact on yield.

[0007] To minimize environmental factors that can cause defects in the workpiece, manufacturing processes and inspections processes are often performed in controlled environments, such as an

enclosed vacuum chamber. Vacuum pressure within the vacuum chamber can be produced using a vacuum pump, such as a turbo molecular pump. These pumps run at high speeds and vibrate, and these vibrations can transfer to the bodies they are attached to within the system. If vibrations are transferred to an inspection system or scanning system, the vibration can reduce accuracy of measurements and produce distorted images. Typical damping systems aim to dampen the connection between the vacuum pump and the vacuum chamber. However, these damping systems may be less effective at high speeds and may be limited to damping along one axis. In order to provide damping along other axes, additional dampers may be used, which increases the complexity and overall mass of the system.

[0008] Therefore, what is needed is an improved damping system for a vacuum pump.

BRIEF SUMMARY OF THE DISCLOSURE

[0009] An embodiment of the present disclosure provides a system. The system may comprise a vacuum chamber, a vacuum pump, and a mass damper. The vacuum chamber may include an internal volume. The vacuum pump may have a pump body, a vacuum inlet in fluid communication with the internal volume, a vacuum outlet, and a rotor assembly disposed within the pump body. The rotor assembly may be configured to rotate about a rotary axis to pump gas from the internal volume of the vacuum chamber out of the vacuum outlet. The mass damper may be connected to the vacuum pump. The mass damper may include a conic portion defined by a converging end and a diverging end. The conic portion may have a cone axis that is coaxial with the rotary axis of the vacuum pump, and a damping mass may be connected to the converging end of the conic portion.

[0010] In some embodiments, the vacuum pump may be a turbo molecular pump. The rotor assembly may be configured to rotate at a speed of at least 50,000 rpm.

[0011] In some embodiments, the vacuum pump may have a vibration frequency of greater than 1 kHz due to rotation of the rotor assembly.

[0012] In some embodiments, the mass damper may further include a base portion connected to the vacuum pump. The base portion may surround the diverging end of the conic portion.

[0013] In some embodiments, an end surface of the base portion may be flush with the diverging end of the conic portion.

[0014] In some embodiments, a plurality of mounting holes may be defined in the base portion. The base portion may be connected to the vacuum pump by a plurality of fasteners extending through the plurality of mounting holes.

[0015] In some embodiments, the plurality of mounting holes may be located at rotationally symmetrical positions relative to the cone axis and are parallel to the cone axis.

[0016] In some embodiments, a plurality of embossed protrusions may be defined on the end surface of the base portion. Each of the plurality of embossed protrusions may surround one of the plurality of mounting holes.

[0017] In some embodiments, a threaded hole may be defined on a planar surface of the damping mass and may be coaxial with the cone axis. The mass damper may further include a tuning mass connected to the damping mass by a fastener received in the threaded hole.

[0018] In some embodiments, the conic portion may be hollow.

[0019] In some embodiments, the mass damper may further include a cylindrical portion. The damping mass may be connected to the converging end of the conic portion by the cylindrical portion.

[0020] In some embodiments, the cylindrical portion may have a diameter that is less than a diameter of the converging end of the conic portion.

[0021] In some embodiments, the damping mass may have a diameter that is greater than a diameter of the cylindrical portion.

[0022] In some embodiments, the mass damper may be rotationally symmetrical relative to the cone axis.

[0023] In some embodiments, the rotary axis of the vacuum pump may be arranged vertically.

[0024] In some embodiments, the rotary axis of the vacuum pump may be arranged horizontally.

[0025] In some embodiments, the rotary axis of the vacuum pump may be arranged at an oblique angle.

[0026] In some embodiments, the vacuum pump may be disposed on top of the vacuum chamber. The mass damper may be disposed on top of the vacuum pump.

[0027] In some embodiments, the vacuum pump may be disposed beneath the vacuum chamber. The mass damper may be disposed beneath the vacuum pump.

[0028] Another embodiment of the present disclosure provides a method. The method may comprise connecting a vacuum inlet of a vacuum pump to a vacuum chamber. The vacuum pump may include a pump body and a rotor assembly disposed within the pump body, and the rotor assembly may be configured to rotate about a rotary axis. The method may further comprise connecting a mass damper to the vacuum pump. The mass damper may include a conic portion defined by a converging end and a diverging end, the conic portion may have a cone axis that is coaxial with the rotary axis of the vacuum pump, and a damping mass may be connected to the converging end of the conic portion. The method may further comprise driving the rotor assembly to rotate within the pump body to pump gas from an internal volume of the vacuum chamber out of a vacuum outlet.

Description

DESCRIPTION OF THE DRAWINGS

[0029] For a fuller understanding of the nature and objects of the disclosure, reference should be made to the following detailed description taken in conjunction with the accompanying drawings, in which:

[0030] FIG. 1 is a side view of a system according to an embodiment of the present disclosure;

[0031] FIG. 2 is a top view of a mass damper of a system according to an embodiment of the present disclosure;

[0032] FIG. 3 is a section view along line A-A of the mass damper of FIG. 2;

[0033] FIG. 4 is a section view along line B-B of the mass damper of FIG. 2;

[0034] FIG. 5 is another section view along line A-A of the mass damper of FIG. 2, where a tuning mass is connected to the damping mass;

[0035] FIG. 6 is another section view along line B-B of the mass damper of FIG. 2, where a tuning mass is connected to the damping mass;

[0036] FIG. 7 is a top perspective view of a mass damper according to another embodiment of the present disclosure;

[0037] FIG. 8 is a bottom perspective view of the mass damper of FIG. 7;

[0038] FIG. 9 is a side view of a system according to another embodiment of the present disclosure, where the vacuum pump is arranged vertically beneath the vacuum chamber;

[0039] FIG. 10 is a side view of a system according to another embodiment of the present disclosure, where the vacuum pump is arranged horizontally next to the vacuum chamber;

[0040] FIG. 11 is a side view of a system according to another embodiment of the present disclosure, where the vacuum pump is arranged at an oblique angle next to the vacuum chamber; and

[0041] FIG. 12 is a flowchart of a method according to an embodiment of the present disclosure.

DETAILED DESCRIPTION OF THE DISCLOSURE

[0042] Although claimed subject matter will be described in terms of certain embodiments, other embodiments, including embodiments that do not provide all of the benefits and features set forth herein, are also within the scope of this disclosure. Various structural, logical, process step, and electronic changes may be made without departing from the scope of the disclosure. Accordingly,

the scope of the disclosure is defined only by reference to the appended claims.

[0043] An embodiment of the present disclosure provides a system **100**. The system **100** may be, for example, an inspection system or a metrology system configured inspect or measure one or more properties of a workpiece. The workpiece may be, for example, a semiconductor wafer, substrate, printed circuit board (PCB), flat panel display (FPD), integrated circuit (IC), or other type of workpiece.

[0044] The system **100** may comprise a vacuum chamber **110**, as shown in FIG. **1**. The vacuum chamber **110** may define an internal volume **111**, in which one or more inspection or metrology processes may be performed on the workpiece. The vacuum chamber **110** may be supported by one or more air mounts **112**, which can reduce vibration transfer from the environment to the vacuum chamber **110**.

[0045] The system **100** may further comprise a vacuum pump **120**. The vacuum pump **120** may have a pump body **122**, a vacuum inlet **123** in fluid communication with the internal volume **111** of the vacuum chamber **110**, a vacuum outlet **124**, and a rotor assembly **125** disposed within the pump body **122**. The rotor assembly **125** may be configured to rotate about a rotary axis **121**. The rotor assembly **125** may include a rotatable shaft driven by a motor and a plurality of rotor blades connected to the shaft. The vacuum pump **120** may further include a plurality of stator blades within the pump body **122**, and the plurality of rotor blades may be configured to rotate relative to the plurality of stator blades. Rotation of the rotor assembly **125** (i.e., the shaft and the plurality of rotor blades) may pump gas from the internal volume **111** of the vacuum chamber **110** (via the vacuum inlet **123**) through the pump body **122** and out of the vacuum outlet **124**. Accordingly, the vacuum pump **120** may be configured to produce a vacuum pressure within the internal volume **111** of the vacuum chamber **110**. In some embodiments, the vacuum pressure may be high vacuum pressure (1×10^{-3} to 1×10^{-9} Torr) or ultra-high vacuum pressure (1×10^{-9} to 1×10^{-12} Torr). The vacuum pump **120** may be connected to the vacuum chamber **110** by a suspension assembly **113**.

[0046] The suspension assembly **113** may include one or more springs and/or dampers configured to reduce transmission of vibrations from the vacuum pump **120** to the vacuum chamber **110**. The suspension assembly **113** may be built-in with the structure of the vacuum pump **120** or may be a separate element provided between the vacuum pump **120** and the vacuum chamber **110**.

[0047] In some embodiments, the vacuum pump **120** may be a turbo molecular pump. The rotor assembly **125** may be configured to rotate at a speed of at least 50,000 rpm to produce the vacuum pressure. At these speeds, the vacuum pump **120** may have a vibration frequency of greater than 1 kHz due to the rotation of the rotor assembly **125**. In some embodiments, the vacuum pump **120** may have a vibration frequency for greater than 2 kHz due to the rotation of the rotor assembly **125**. In an instance, the vibration frequency may be 3.2 kHz or more. At these vibration frequencies, the suspension assembly **113** may be unable to effectively reduce transmission of vibrations from the vacuum pump **120** to the vacuum chamber **110**. In addition, due to the rotation of the rotor assembly **125**, the vibration may induce movement of the vacuum pump **120** along three axes (i.e., the rotary axis **121** and two axes orthogonal to the rotary axis **121**), which may be difficult for the suspension assembly **113** to compensate for.

[0048] The system **100** may further comprise a mass damper **130**. The mass damper **130** may be connected to the vacuum pump **120** and may reduce the vibration in the system **100** generated by the vacuum pump **120**. Referring to FIGS. **2-8**, the mass damper **130** may include a conic portion **132** defined by a converging end **132a** and a diverging end **132b**. The conic portion **132** may be a frustoconical shape having a curved or tapered profile from the converging end **132a** to the diverging end **132b**. In some embodiments, the conic portion **132** may be hollow. For example, the conic portion **132** may include a hollow volume **132c**, which can be open to the diverging end **132b**. The conic portion **132** may have a cone axis **131** that is coaxial with the rotary axis **121** of the vacuum pump **120**. The mass damper **130** may be rotationally symmetrical relative to the cone

axis **131**. The mass damper **130** may further include a damping mass **133** connected to the converging end **132a** of the conic portion **132**. In some embodiments, the damping mass **133** may be removably connected to the conic portion **132**. The damping mass **133** may have a cylindrical shape. The damping mass **133** may be integrally formed with the conic portion **132**, or the damping mass **133** may be a separate element connected to the conic portion **132**.

[0049] In some embodiments, the mass damper **130** may further include a cylindrical portion **134**. The damping mass **133** may be connected to the converging end **132a** of the conic portion **132** by the cylindrical portion **134**. As shown in FIG. 3 and FIG. 4, the cylindrical portion **134** may have a diameter that is less than a diameter of the converging end **132a** of the conic portion **132**. In some embodiments, the damping mass **133** may have a diameter that is greater than the diameter of the cylindrical portion **134**. In some embodiments, the damping mass **133** may be directly connected to the converging end **132a** of the conic portion **132**, without a cylindrical portion **134** disposed therebetween (as shown in FIG. 7 and FIG. 8).

[0050] In some embodiments, the mass damper **130** may further include a base portion **135**. The base portion **135** may surround the diverging end **132b** of the conic portion **132**. For example, an end surface **135a** of the base portion **135** may be flush with the diverging end **132b** of the conic portion. The base portion **135** may connect the mass damper **130** to the vacuum pump **120**. For example, the end surface **135a** of the base portion **135** may be disposed on the pump body **122** of the vacuum pump **120**. The shape of the base portion **135** may depend on the shape of the pump body **122**. For example, as shown in FIG. 2, the base portion **135** may have a substantially square shape. However, in some embodiments, the base portion **135** may have a substantially circular shape or other shapes. A plurality of mounting holes **136** may be defined in the base portion **135**, and the base portion **135** may be connected to the vacuum pump **120** by a plurality of fasteners **126** extending through the plurality of mounting holes **136**. The number and arrangement of the plurality of mounting holes **136** of the base portion may depend on the corresponding structure of the pump body **122** of the vacuum pump **120**. The plurality of mounting holes **136** may be located at rotationally symmetrical positions relative to the cone axis **131** (as shown in FIG. 2) and may be parallel to the cone axis **131** (as shown in FIG. 4).

[0051] In some embodiments, a plurality of embossed protrusions **135b** may extend from the end surface **135a** of the base portion **135**. The plurality of embossed protrusions **135b** may be ring-shaped and surround the plurality of mounting holes **136**. The plurality of embossed protrusions **135b** may be received in corresponding recesses of the vacuum pump **120**, so as to align the mass damper with the vacuum pump **120** for insertion of the plurality of fasteners **126** through the plurality of mounting holes **136** and precisely define the boundary conditions of the mass damper **130**. With the plurality of embossed protrusions **135b** received in the recesses of the vacuum pump **120**, relative rotation between the mass damper **130** and the vacuum pump **120** can be prevented or minimized.

[0052] In some embodiments, a threaded hole **137** may be defined on a planar surface **133a** of the damping mass **133**. The threaded hole **137** may be coaxial with the cone axis **131**. The mass damper **130** may further include a tuning mass **138** connected to the damping mass **133** by a fastener **139** receiving in the threaded hole **137**, as shown in FIG. 5 and FIG. 6. The tuning mass **138** may have a cylindrical shape. The tuning mass **138** may be configured to increase the overall mass of the mass damper **130** and may adjust the location of the center of mass of the mass damper **130** to further reduce vibrations generated by the vacuum pump **120**. In some embodiments, the tuning mass **138** may have the same dimensions as the damping mass **133**. Alternatively, the tuning mass **138** may have different dimensions (e.g., diameter and/or thickness) for fine adjustment of the overall mass and the center of mass of the mass damper **130**. For example, a thicker tuning mass **138** may move the center of mass of the mass damper **130** along the cone axis **131** compared to a larger diameter tuning mass **138** of the same mass.

[0053] Based on the rotationally symmetrical design of the mass damper **130** and its specifically

tuned overall mass and center of mass, the mass damper **130** can reduce the vibration in the system **100** generated by the vacuum pump **120** in various orientations. For example, the vacuum pump **120** can be arranged on top of the vacuum chamber **110** with the rotary axis **121** arranged vertically, and the mass damper **130** can be arranged on top of the vacuum pump **120**, as shown in FIG. **1**. Alternatively, the vacuum pump can be arranged beneath the vacuum chamber **110** with the rotary axis **121** arranged vertically, and the mass damper **130** can be disposed beneath the vacuum pump **120**, as shown in FIG. **9**. In some embodiments, the vacuum pump **120** can be arranged adjacent to the vacuum chamber **110**, with different orientations of the rotary axis **121**. For example, the rotary axis **121** may be arranged horizontally (as shown in FIG. **10**) or at an oblique angle θ (as shown in FIG. **11**). The oblique angle θ may be an angle between 0° and 90° . In an instance, the oblique angle Γ may be 45° . In any arrangement of the vacuum pump **120** relative to the vacuum chamber **110** and arrangement of the rotary axis **121**, the cone axis **131** of the mass damper **130** may be coaxial with the rotary axis **121**, such that the mass damper **130** provides three-axis vibration damping for the vacuum pump **120**.

[0054] With the system **100**, the vibrations generated by the vacuum pump **120** can be reduced (e.g., by 40% to 90%) by simply attaching the mass damper **130** to the vacuum pump **120**. While the structure of the mass damper **130** is designed for specific operating speeds of the vacuum pump **120**, the mass damper **130** can be tuned for different operating speeds by connecting an additional tuning mass **138** to the damping mass **133** of the mass damper **130**. Accordingly, the mass damper **130** can effectively reduce transmission of vibrations from the vacuum pump **120** to the vacuum chamber **110**, which can improve accuracy of measurements and image quality of inspection and metrology systems provided in the vacuum chamber **110**.

[0055] Another embodiment of the present disclosure provides a method **200**. As shown in FIG. **12**, the method **200** may comprise the following steps.

[0056] At step **210**, a vacuum inlet of a vacuum pump is connected to a vacuum chamber. The vacuum pump may include a pump body and a rotor assembly disposed within the pump body and configured to rotate about a rotary axis.

[0057] At step **220**, a mass damper is connected to the vacuum pump. The mass damper may include a conic portion defined by a converging end and a diverging end. The conic portion may have a cone axis that is coaxial with the rotary axis of the vacuum pump. A damping mass may be connected to the converging end of the conic portion. The structure of the damping mass may be designed to reduce vibrations generated by the vacuum pump at specific operating speeds of the vacuum pump.

[0058] In some embodiments, the first vibration resonance frequency of the mass damper may be measured and compared to the running frequency of the vacuum pump. If there is a large difference between the frequencies, the damping mass can be removed or additional tuning masses can be added or removed from the mass damper. The tuning mass may be configured to modify the overall mass and the center of mass of the mass damper, so as to reduce vibrations generated by the vacuum pump at different operating speeds. For example, an iterative process of adding or removing tuning masses to the mass damper and measuring the vibrations transmitted to the system can be performed until the first resonance of the mass damper matches the vacuum pump running frequency.

[0059] At step **230**, the rotor assembly is driven to rotate within the pump body to pump gas from an internal volume of the vacuum chamber out of a vacuum outlet.

[0060] With the method **200**, the vibrations generated by the vacuum pump can be reduced (e.g., by 60% to 90%) by simply attaching the mass damper to the vacuum pump. While the structure of the mass damper is designed for specific operating speeds of the vacuum pump, the mass damper can be tuned for different operating speeds by connecting an additional tuning mass to the damping mass of the mass damper. Accordingly, the mass damper can effectively reduce transmission of vibrations from the vacuum pump to the vacuum chamber, which can improve accuracy of

measurements and image quality of inspection and metrology systems provided in the vacuum chamber.

[0061] Although the present disclosure has been described with respect to one or more particular embodiments, it will be understood that other embodiments of the present disclosure may be made without departing from the scope of the present disclosure. Hence, the present disclosure is deemed limited only by the appended claims and the reasonable interpretation thereof.

Claims

1. A system comprising: a vacuum chamber including an internal volume; a vacuum pump having a pump body, a vacuum inlet in fluid communication with the internal volume, a vacuum outlet, and a rotor assembly disposed within the pump body and configured to rotate about a rotary axis to pump gas from the internal volume of the vacuum chamber out of the vacuum outlet; and a mass damper connected to the vacuum pump, wherein the mass damper includes a conic portion defined by a converging end and a diverging end, the conic portion having a cone axis that is coaxial with the rotary axis of the vacuum pump, and a damping mass connected to the converging end of the conic portion.
2. The system of claim 1, wherein the vacuum pump is a turbo molecular pump, and the rotor assembly is configured to rotate at a speed of at least 50,000 rpm.
3. The system of claim 1, wherein the vacuum pump has a vibration frequency of greater than 1 kHz due to rotation of the rotor assembly.
4. The system of claim 1, wherein the mass damper further includes a base portion connected to the vacuum pump, the base portion surrounding the diverging end of the conic portion.
5. The system of claim 4, wherein an end surface of the base portion is flush with the diverging end of the conic portion.
6. The system of claim 4, wherein a plurality of mounting holes are defined in the base portion, and the base portion is connected to the vacuum pump by a plurality of fasteners extending through the plurality of mounting holes.
7. The system of claim 6, wherein the plurality of mounting holes are located at rotationally symmetrical positions relative to the cone axis and are parallel to the cone axis.
8. The system of claim 6, wherein a plurality of embossed protrusions are defined on an end surface of the base portion, and each of the plurality of embossed protrusions surrounds one of the plurality of mounting holes.
9. The system of claim 1, wherein a threaded hole is defined on a planar surface of the damping mass and is coaxial with the cone axis, and the mass damper further includes a tuning mass connected to the damping mass by a fastener received in the threaded hole.
10. The system of claim 1, wherein the conic portion is hollow.
11. The system of claim 1, wherein the mass damper further includes a cylindrical portion, and the damping mass is connected to the converging end of the conic portion by the cylindrical portion.
12. The system of claim 11, wherein the cylindrical portion has a diameter that is less than a diameter of the converging end of the conic portion.
13. The system of claim 11, wherein the damping mass has a diameter that is greater than a diameter of the cylindrical portion.
14. The system of claim 1, wherein the mass damper is rotationally symmetrical relative to the cone axis.
15. The system of claim 1, wherein the rotary axis of the vacuum pump is arranged vertically.
16. The system of claim 1, wherein the rotary axis of the vacuum pump is arranged horizontally.
17. The system of claim 1, wherein the rotary axis of the vacuum pump is arranged at an oblique angle.
18. The system of claim 1, wherein the vacuum pump is disposed on top of the vacuum chamber,

and the mass damper is disposed on top of the vacuum pump.

19. The system of claim 1, wherein the vacuum pump is disposed beneath the vacuum chamber, and the mass damper is disposed beneath the vacuum pump.

20. A method comprising: connecting a vacuum inlet of a vacuum pump to a vacuum chamber, wherein the vacuum pump includes a pump body and a rotor assembly disposed within the pump body and configured to rotate about a rotary axis; connecting a mass damper to the vacuum pump, wherein the mass damper includes a conic portion defined by a converging end and a diverging end, the conic portion having a cone axis that is coaxial with the rotary axis of the vacuum pump, and a damping mass connected to the converging end of the conic portion; and driving the rotor assembly to rotate within the pump body to pump gas from an internal volume of the vacuum chamber out of a vacuum outlet.
